Machine Learning Based Clinical Trajectory Prediction in Emergency Care

Research Questions & Paper Outline

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Table of Contents

Research Question 1 (RQ1)

Feature Classification

Can features identified via SHAP be systematically classified into **predictive**, **redundant**, or **harmful** categories based on their causal impact when removed?

Where answered: Results 3.2 (Feature Ablation), Table 1

Key evidence:

- Triage acuity (predictive): removal reduces sensitivity by 7-8 pp
- EMS arrival (redundant): removal changes sensitivity by ± 0.4 pp in tree-based models
- Charlson index (harmful): removal improves sensitivity by $+16.0$ pp (LightGBM)

Research Question 2 (RQ2)

Integrative Outcome Prediction

Using only admission-available variables, can ML models predict three interrelated outcomes (ED-LOS, admission, Hosp-LOS) with stable performance suitable for real-time deployment?

Where answered: Results 3.3-3.7, Tables 2-3

Key evidence:

- LightGBM: 82.0% sensitivity for prolonged ED stay (95% CI: 80.7-85.7)
- XGBoost: AUROC 0.809 for admission (95% CI: 0.802-0.817)
- Random Forest: 73.5% sensitivity for prolonged Hosp-LOS (95% CI: 70.9-76.1)
- Confidence interval widths: 2.8% to 5.2%

Research Question 3 (RQ3)

Fairness and Interpretability

Do prediction models exhibit demographic disparities, and can these be mitigated via threshold adjustment without removing demographic features?

Where answered: Results 3.6-3.9, Tables 5-8

Key evidence:

- SHAP stability across models: 75-79% Jaccard similarity
- Baseline racial disparity: 9.1 percentage points
- Post-adjustment disparity: 0.5 percentage points
- Threshold reduction: 15% for Black, 12% for Hispanic/Other

Abstract

Target: $<$200 words

Structure:

1. Background (1 sentence)
2. Methods (2 sentences)
3. Results (3-4 sentences with key numbers)
4. Conclusion (3 sentences)

Introduction (1.5-2 pages)

1. **1.1 Problem statement**
   - ED crowding and 37% mortality risk for prolonged stay
2. **1.2 Literature gaps (4 gaps)**
   - Fragmented outcomes (single-outcome models)
   - Late-feature dependency
   - Single-model SHAP only
   - No feature classification framework
3. **1.3 Research questions (RQ1, RQ2, RQ3)**
4. **1.4 Contributions**

Materials and Methods (3-4 pages)

2.1 Study design & data source
2.2 Inclusion criteria
2.3 Feature classification (RQ1)
2.4 Missing data handling
2.5 Prediction tasks (RQ2)
2.6 Data split
2.7 ML models (12 algorithms)
2.8 Evaluation metrics & CIs
2.9 Statistical analysis
2.10 Interpretability & fairness (RQ3)

Will be allowed if I don't include the 12 Machine learning models trained and just use only 4? Am just being curious because a model like SVM was never used in the research.

Results (4-5 pages)

3.1 Cohort characteristics
3.2 Feature ablation (RQ1) → Table 1
3.3 Full 12-model performance (RQ2)
3.4 Severity-stratified → Table 4
3.5 Model calibration → Table 5
3.6 SHAP stability (RQ3) → Table 6
3.7 Model selection → Table 3
3.8 Demographic disparity (RQ3) → Tables 7-8

Discussion (2-3 pages)

1. **4.1** Summary of findings by RQ
2. **4.2** Discussion of RQ1 (feature classification)
3. **4.3** Discussion of RQ2 (integrative prediction)
4. **4.4** Discussion of RQ3 (fairness & interpretability)
5. **4.5** Comparison with prior work
6. **4.6** Practical implications
7. **4.7** Limitations (5 limitations)

Conclusion and Required Sections

Conclusion (0.5 page)

- Answer each RQ briefly
- Summarize key numbers
- State broader implications

Required Sections

- Conflict of Interest
- Author Contributions
- Funding
- Ethics Statement
- Acknowledgments
- Data Availability
- Supplementary Materials
- References